Econ 272/Mgtecon 607 - Section 4

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Outline

1. Announcements
2. Directed Acyclic Graphs (DAGs)
3. Practice Problems
4. (Evolution of) Views On Applied Econometrics

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2. Directed Acyclic Graphs (DAGs)
3. Practice Problems
4. (Evolution of) Views On Applied Econometrics

Final Exam

- Final Exam: Wednesday June 11, 8:30-11:30am
- Remember to email me about whether you wish to take the final exam, and if you require any special accommodations for it
  - Unless you are an Econ PhD, in which case you have no choice :)
  - If you are a predoc and wish to waive the class next year, you must take the exam as well

1. Announcements

2. Directed Acyclic Graphs (DAGs)

3. Practice Problems

4. (Evolution of) Views On Applied Econometrics

Discussion Questions - DAGs

1. What is a Directed Acyclical Graph and what is its use?
2. What is the front door criterion?
3. What is the backdoor criterion?
4. What is M-bias?

Structural Equation Models $\iff$ DAGs

Consider a simple randomized experiment.

- **Structural Equation:**

$$W = f_1(\varepsilon_1) \qquad Y = f_2(W, \varepsilon_2)$$

random

- **DAG:**

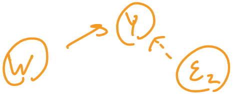
$$\begin{tikzcd}
\varepsilon_1 \arrow[r, dashed] & W \arrow[d] \\
\varepsilon_2 \arrow[r, dotted] & Y
\end{tikzcd}$$

Why DAGs? Help (visually) argue for identification

Terminology C → A → B

- **Nodes:** Quantities / variables (measured or unmeasured) (A, B, C)
- **Parent/Child:** Immediate predecessor / successor (A to B, C to A)
- **Ancestor:** Not immediate predecessor (C to B)
- **(Dashed) Arrows:** (Unobserved) links (ε in prev. slide)
- **Directed Path:** Path with all arrows in same direction (C → A → B)

- **Collider:** Arrows coming in, but not out (on a path)

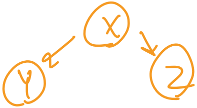


Y is a collider

- **Mediator:** 1 arrow in, 1 arrow out (on a path)

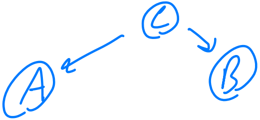
W is a mediator

- **Confounder:** Arrows going out, but not in (on a path)



X is a confounder

- **Backdoor Path:** Starts → A, ends → B



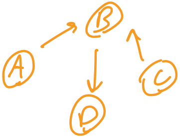
- **Blocked Path:** Set of nodes Z, blocks path if $\exists z$ on path s.t.

  ① z is a noncollider that has been conditioned on

  $A \to B \to C$ blocked conditioned on B

  OR

  ③ z is a collider not conditioned on and all its descendants not in Z,



B blocks $A \to B \leftarrow C$

- **Open Path:** Any path that is not blocked

  Mediators, confounders generally will not block (without conditioning)

Blocking and Conditioning

Conditioning on a variable reverses blocking rules

including in regression model

$$A \rightarrow B \rightarrow C$$

Unblocked

Blocked conditional on B

$$\begin{tikzcd} & B & \\ A \arrow[ur] & & C \arrow[ul] \end{tikzcd}$$

Blocked by B

Unblocked conditional on B

Including extra variables in regression (conditioning) may be necessary for identification.

Extension: d-Separation

- **Definition:** A set of variables d-separates 2 other variables if it conditionally blocks all paths between them.

- **Implication:** If A is d-separated from B by set Z_1, then

  $A \perp B \mid Z_1$

  conditional independence

Identification Criteria (for causal effect of A on B)

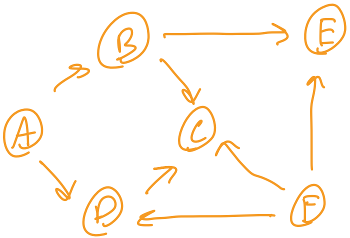
- Back Door Criterion: Can we block all paths by conditioning on some set of nodes? If yes, causal effect of $A \to B$ identified (conditioned on the set)

- Front Door Criterion: Set of nodes Z_1 satisfies:
  1. Z_1 intercepts all directed paths from $A \to B$
  2. No unblocked backdoor path from $A \to Z_1$
  3. All backdoor paths from $Z_1 \to B$ blocked by A

  If all 3 hold, then causal effect $A \to B$ identified (conditioned on Z_1)

Back Door Criterion - Example

Can we ID $D \to E$?



What if A unobserved ?

→ cannot condition on it, but $\{B, F\}$ still works

Paths with unobserved nodes still need to be blocked !

Backdoor paths:

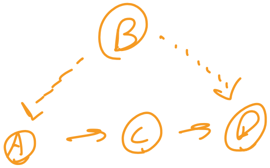
- $D \leftarrow A \leftarrow B \rightarrow E$
  blocked by conditioning on $\{A\}$, $\{B\}$, $\{A, B\}$
- $D \leftarrow F \leftarrow E$
  blocked by conditioning on $\{F\}$
- $D \leftarrow F \rightarrow C \leftarrow B \rightarrow E$
  blocked by not conditioning on $\{C\}$ (or any of its descendants) → collider
- $D \leftarrow A \rightarrow B \rightarrow C \leftarrow F \rightarrow E$
  same as above

Can block all of above by conditioning on: $\{A, F\}$, $\{B, F\}$, $\{A, B, F\}$

generally prefer conditioning on fewest variables possible

Front Door Criterion - Example

Can we ID $A \to D$



Directed paths!

- $A \to C \to D$

  blocked conditioned on $\{C\}$

1. Only 1 directed path: $Z_1 = \{C\}$
2. No backdoor path from $A \to C$
3. 1 backdoor path from $C \to D$:

   $C \leftarrow A \leftarrow B \to D$

   blocked conditioned on A

All 3 criteria satisfied by conditioning on $\{C\}$.

~~M-bias~~ Collider Bias

Conditioning on a collider can unblock paths and introduce bias

M-bias: Special case of mistaking collider for confounder.

Ex. "general" bias

Goal: impact of attendance → good grades

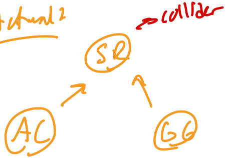
Method: Run survey

Problem: Good students more likely to respond

Ideal:

SR ideally independent of both

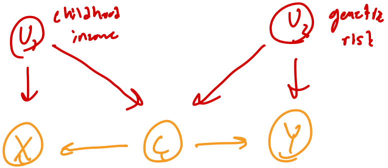
Actual:



Ex. M-bias

Goal: impact of education → diabetes

Think: confounded by mother's diabetes status



But, it is really a consequence of unobserved confounding and is therefore a collider; conditioning will open a backdoor path from $X \to Y$, introduce bias (no ID)

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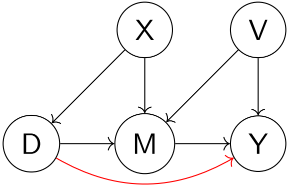
4. (Evolution of) Views On Applied Econometrics

(a) Can we identify a causal effect of D on Y?

(b) What if X and V are unobserved?



(a) Backdoor criterion

Backdoor paths:

$D \leftarrow X \rightarrow M \rightarrow Y$ → blocked by $\{X\}, \{M\}, \{X, M\}$

$D \leftarrow X \rightarrow M \leftarrow V \leftarrow Y$

→ M is a collider, blocked by not conditioning on M or its descendants

ID conditional on $\{X\}$

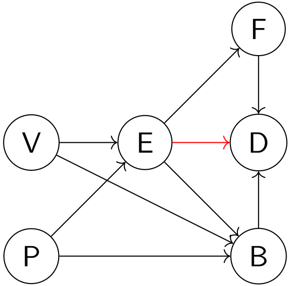
(b) Cannot condition on $\{X\}$, must condition on $\{M\}$ to block 1st path.

But, that requires conditioning on X or V to block second path (conditioning on collider unblocks path)

So, backdoor fails.

Front door also fails: cannot block backdoor paths $D \rightarrow M$, $M \rightarrow Y$. $\boxed{\text{So, no ID}}$

Can we identify a causal effect of E on D?



Backdoor criterion

Backdoor paths:

$E \leftarrow V \rightarrow B \rightarrow D$

$E \leftarrow P \rightarrow B \rightarrow D$

Blocked by:

$\{V\}, \{B\}, \{V, B\}$

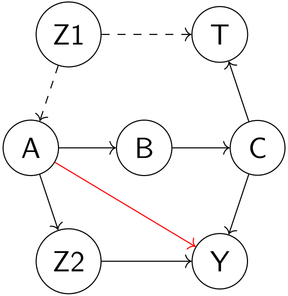
$\{P\}, \{B\}, \{P, B\}$

ID by conditioning on:

$\{B\}, \{V, P\}, \{B, P\}, \{V, B, P\}$

favor smallest

Can we identify a causal effect of A on Y?



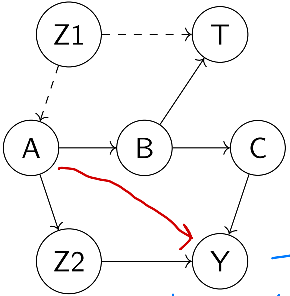
Backdoor paths:

$A \leftarrow Z_1 \rightarrow T \leftarrow C \rightarrow Y$

T is a collider, so the path is blocked with no conditioning.

DAG 3(b)

What if we add an arrow from B to T and delete the arrow from C to T. Can we use the frontdoor criterion for identification?



① Directed paths:

$A \to B \to C \to Y$ — Blocked by: $\{B\}$, $\{C\}$, $\{B, C\}$

$A \to Z_2 \to Y$ — $\{Z_2\}$

Candidates: $\{B, Z_2\}$, $\{C, Z_2\}$, $\{B, C, Z_2\}$

② No unblocked backdoor from A to Z → trivially satisfied, no backdoor at all

③ A blocks all backdoor from Z to Y

$B \leftarrow A \to Z_2 \to Y$

$C \leftarrow B \leftarrow A \to Z_2 \to Y$

$C \leftarrow B \to T \leftarrow Z_1 \to A \to Z_2 \to Y$

$Z_2 \leftarrow A \to B \to C \to Y$

$Z_2 \leftarrow A \leftarrow Z_1 \to T \leftarrow B \to C \to Y$

All blocked by A

→ T collider, A not descendant

So, ID conditional on any of:

$\{B, Z_2\}$, $\{C, Z_2\}$, $\{B, C, Z_2\}$

favor smaller

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4. (Evolution of) Views On Applied Econometrics

Big idea: Econometrics = mathematical statistics

Specify a single model, estimate parameters once

No model uncertainty, only sampling variation

Main takeaway: Applied econometricians "robots", theoretical econometricians important

Far from reality of empirical work

Leamer says: We actually test lots of models, but report estimates as if we only test one

Big idea: Modeler specifies algorithm, which is fully specified, for model selection.

CIs constructed accordingly.

Main takeaway: Still unrealistic; never have fully pre-specified algorithm.

Big idea: Start with a general model that fits data well, then iteratively drop insignificant covariates

Primarily in context of time-series models

Main takeaway: How to do the simplification? Unclear how good proposed algos are in practice.

End models may not be causally useful.

Simplification art, not science.

Big idea: We have to search over models, and cannot totally pre-specify, so just be honest.
Systematically test + explore multiple specifications.
Take extra uncertainty into account.

Main takeaway: Hard to specify and defend why your ex-ante selected specification set captures the relevant models.

Big idea: Specify form/amount of dependence of parameters on unobserved quantities.

Consider "reasonable" ranges of the unobserved value and observe induced change in ATE point estimate.

Overall Main takeaway: Randomized experiments > observational

But, when done well, observational hugely influential

Clearly some need to worry about specification searching, but unclear what the "right" thing is.